NED University of Engineering and Technology, Karachi -75270, Pakistan

DEPARTMENT OF MECHANICAL ENGINEERING

AI & Internet of Things (ME-446)

Complex Engineering Problem - Fall Semester 2026

Mapped CLO: CLO-2 (C3) - PLO-2 Problem Analysis	Max. Marks: 14
Group Size: Maximum 4 students per group	**Dataset:** Open-choice and unique for every group
Announcement Date: 16-Sept-2026	**Submission Date:** 04-Nov-2026

A. Problem Statement

Mechanical engineering systems increasingly generate data from experiments, sensors, simulations, maintenance records, manufacturing processes, energy systems and field operations. In practice, such datasets are rarely clean or uniform, and the suitability of an AI/ML algorithm depends on the engineering objective, data quality, feature characteristics, target variable and required prediction or decision. Therefore, there is no single prescribed preprocessing method or machine-learning solution.

For this CEP, each student group shall independently select a different open-choice dataset from a publicly accessible source and define a meaningful mechanical engineering problem involving prediction, classification, estimation, diagnosis, condition monitoring, or decision support. Each group shall preprocess and prepare the selected dataset using appropriate techniques, **apply**, and compare multiple AI/ML algorithms, and visualize the engineering data and key results through an interactive dashboard. The group shall also evaluate the performance of the developed models and interpret the findings in the context of the selected mechanical engineering application.

Important: There is no fixed list of datasets. Every group must work on a different dataset. Dataset approval by the course teacher is mandatory before analysis begins.

B. Scope of Work / Tasks

I. Open Dataset Selection and Engineering Problem Formulation

1. Select one publicly accessible dataset relevant to mechanical engineering, such as manufacturing, condition monitoring, vibration, thermal/HVAC systems, energy, renewable energy, automotive/EV systems, materials, fluids, maintenance, or a similar engineering application.
2. Define the engineering problem, input variables/features, target/output variable, and the significance of the selected variables. Clearly state whether the task is regression, or classification, or another justified AI/ML problem.
3. Register the dataset with the course teacher. Once approved, the same dataset cannot be used by any other group, even with a different target variable or subset, unless explicitly permitted by the teacher.

II. Data Preprocessing, Visualization and Dashboard

4. Audit the raw data for data types, duplicate records, missing values, outliers, inconsistent units, categorical variables, class imbalance where applicable, and possible data leakage.
5. Apply justified preprocessing and, where appropriate, feature engineering, normalization/scaling, encoding, outlier treatment, imbalance handling or dimensionality reduction. Retain and submit both the raw and cleaned datasets.
6. Generate at least three informative visualizations and extract three to five engineering insights. Develop an interactive and professional dashboard containing suitable KPIs, filters and views that support understanding of the dataset and the engineering problem.

III. AI/ML Algorithm Application and Model Comparison

7. Apply at least three appropriate AI/ML algorithms or modeling approaches. The choice of algorithms must be justified according to the engineering problem, target variable and dataset characteristics.
8. Use an appropriate training/testing strategy and prevent information leakage between datasets. For time-series data, use a chronological split unless another method is technically justified.
9. Compare the implemented models using task-appropriate evaluation metrics, explain the observed errors and limitations, and select the preferred model on the basis of evidence rather than accuracy alone. Demonstrate the final model on at least two previously unseen input cases.

IV. Engineering Interpretation and Reporting

10. Interpret the behavior and outputs of the selected model in engineering terms. Where technically applicable, examine important features/variables and discuss engineering plausibility, limitations, possible causes of prediction error, and practical recommendations arising from the analysis.

C. Constraints / Assumptions

- The dataset must be relevant to a mechanical engineering system or process and must be obtained from a public/open source. Generic classroom datasets with no engineering context are not acceptable.
- Every group must use a different dataset. Dataset selection is subject to teacher approval on a first-approved basis.
- The dataset source, access link/reference, date accessed, and any stated license or usage condition must be cited. The original raw dataset must be submitted unchanged.
- The selected dataset should contain sufficient engineering variables and observations to support meaningful preprocessing, visualization, model training and comparison; trivial datasets that do not allow engineering interpretation are not acceptable.
- All assumptions, data exclusions, transformations and feature-engineering decisions must be documented and justified. Target leakage, unjustified deletion of difficult records, or manipulation of results is not permitted.
- All AI/ML outputs must be checked for engineering plausibility. A high numerical score alone is not sufficient; the group must discuss model limitations, generalization and the consequences of incorrect predictions in the selected application.

D. Computational / Modern Tool Requirements

- Python in Jupyter Notebook or Google Colab (e.g., NumPy, pandas, scikit-learn and suitable visualization/modeling libraries).
- Tableau Public for the interactive engineering dashboard, unless an alternative visualization platform is approved by the course teacher.

E. Expected Outcomes / Deliverables

- Technical report in digital and hardcopy form, including engineering problem definition, dataset source, preprocessing decisions, visualizations, dashboard, AI/ML model implementation and comparison, engineering interpretation, limitations and recommendations.
- Raw dataset and cleaned/transformed dataset in CSV or an equivalent machine-readable format.

- Jupyter Notebook/Colab file containing reproducible code for data preprocessing, visualization, AI/ML model training, validation, evaluation and prediction.
- Tableau workbook (or approved equivalent) containing an interactive dashboard with meaningful KPIs/filters.
- Dashboard files and evidence of the dashboard demonstration, including key visualizations, KPIs, filters, model-related outputs and engineering insights.
- Presentation and a concise poster in accordance with the guidelines.

F. CEP Mapping with CLO

CLO No.	CLO Statement	Taxonomy Level	Mapped PLO
2	Apply appropriate AI/ML algorithms and data preprocessing techniques on engineering datasets.	C3	PLO-2 Problem Analysis

G. Complex Engineering Problem Attributes

Attribute	How the CEP satisfies the attribute
Range of conflicting requirements	The group must balance competing requirements such as data cleaning versus information loss, model performance versus interpretability, model complexity versus computational cost, and fit to training data versus robustness/generalization on unseen data.
Depth of analysis required	No single preprocessing method, feature set or AI/ML algorithm is prescribed. Students must analyze the dataset, select appropriate preprocessing steps, compare alternative models and justify the final approach using quantitative evidence and engineering reasoning.
Depth of knowledge required	The work requires integration of mechanical engineering domain knowledge with statistics/data analysis, data preprocessing, feature handling, AI/ML algorithms, model evaluation and visualization/dashboard development.
Familiarity of issues	Each group works on a different externally sourced dataset and a different engineering context; therefore, the data characteristics, target variable, preprocessing issues and suitable model choices extend beyond fixed classroom examples.
Consequences	Incorrect preprocessing, inappropriate algorithm selection or unreliable predictions may lead to poor maintenance, quality, energy, operational or condition-monitoring decisions, with possible economic, reliability, efficiency, safety or sustainability consequences.
Interdependence	The problem contains interdependent stages: engineering problem formulation, dataset understanding, preprocessing, feature preparation, visualization/dashboard development, AI/ML model application, validation, comparison and engineering interpretation. Decisions made at one stage directly affect the reliability of later stages.

H. Rubric for Assessment of Complex Engineering Problem

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Criterion	Poor (0)	Average (0.5)	Good (1.0)	Very Good (1.5)	Excellent (2.0)
1. Dataset selection and engineering problem formulation	Dataset is not approved/relevant; the engineering problem, inputs or target are unclear.	Dataset is approved, but problem formulation is incomplete or weakly connected to mechanical engineering.	Engineering objective, features and target are defined with basic physical relevance.	Problem is clearly formulated; variables and target are technically justified and appropriate for AI/ML analysis.	A unique, relevant dataset is selected and the engineering problem is formulated rigorously, with clear physical interpretation of inputs, target and intended engineering decision/output.
2. Data preprocessing and feature preparation	Preprocessing is absent or contains major errors that invalidate the analysis.	Only limited cleaning is performed; important issues such as missing data, outliers, encoding, scaling or leakage are not properly handled.	Basic preprocessing is correctly implemented, but some choices are weakly justified or incomplete.	Preprocessing and feature handling are correctly implemented and justified; data leakage is avoided and raw/cleaned data are retained.	Preprocessing is systematic, reproducible and technically justified; relevant data-quality issues, transformations and feature preparation are handled appropriately for the selected algorithms and engineering problem.
3. Data visualization, dashboard and engineering insights	Required visualizations/dashboard are absent or do not communicate useful information.	Some visuals are produced, but the dashboard is incomplete, weakly organized or lacks meaningful KPIs/filters.	A functional dashboard contains relevant visuals and some valid engineering insights.	Interactive dashboard contains suitable KPIs, filters and informative visualizations; 3-5 engineering insights are clearly explained.	Professional and interactive dashboard integrates meaningful KPIs, filters and strong visual analysis; insights are data-supported and clearly connected to the engineering problem and model development.
4. AI/ML algorithm implementation	No working AI/ML model is produced, or implementation errors invalidate the results.	Only one model is attempted, or model implementation contains major technical errors.	At least two appropriate models are implemented with a basically correct workflow.	At least three appropriate AI/ML models are correctly implemented; algorithm choices are reasonably justified.	At least three appropriate AI/ML algorithms are implemented systematically in a reproducible pipeline, with model choices clearly justified from the target variable, data characteristics and engineering objective.
5. Model validation, comparison and final selection	Training/testing or evaluation is absent or invalid.	Validation is weak or inappropriate; metrics are poorly selected and leakage/generalization issues are not addressed.	A valid train/test strategy and basic model comparison are used with generally suitable metrics.	A justified validation strategy is used; leakage is avoided, and models are compared using task-appropriate metrics with discussion of errors and limitations.	Model selection is evidence-based using appropriate validation and metrics; robustness/generalization is considered, and the preferred model is demonstrated on at least two previously unseen input cases.
6. Engineering interpretation, limitations and recommendations	Results are not interpreted in engineering terms or are physically implausible.	Interpretation is mostly descriptive and weakly linked to the selected engineering application.	Key results are related to the engineering context, with basic discussion of limitations.	Model behavior, important variables, engineering plausibility, limitations and practical implications are clearly discussed.	Results are interpreted with strong engineering reasoning; feature effects, error causes, consequences of incorrect predictions and defensible practical recommendations are critically discussed.
7. Reproducibility, technical report and communication	Submission is incomplete; results cannot be reproduced or explained.	Report/code/dashboard are poorly organized or important deliverables are missing.	Report, code and supporting files are adequate; results are generally reproducible and communicated with reasonable clarity.	Clear technical report, reproducible notebook, datasets, dashboard and presentation/poster are provided and coherently organized.	Professional, fully reproducible submission integrates raw/cleaned data, code, dashboard and technical communication; assumptions, evidence, conclusions and limitations are concise, traceable and well documented.
TOTAL	**Marks Obtained**				**/ 14**

Group No.: _____ Teacher Signature: _____

Date: _____